

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI, KARNATAKA



A REPORT ON THE 15-WEEK INTERNSHIP

carried out at

**Vstand4U Technologies Pvt. Ltd.
(Vstand4U Solutions)**

Submitted by
Megharaj P
USN: 4PM22IS029

In partial fulfillment of the requirements for the award of

BACHELOR OF ENGINEERING
in
Information Science and Engineering

Project Title
**FoodShare: A Food Redistribution Web
Application**

Internal Guide
Mr. Vinay SK
Assistant Professor
Department of ISE
PESITM, Shivamogga

External Guide
Industry Mentor
Vstand4U Solutions
Bengaluru

**PES INSTITUTE OF TECHNOLOGY AND
MANAGEMENT**

SHIVAMOGGA – 577204

2025–2026

Institute Certificate

PES INSTITUTE OF TECHNOLOGY AND MANAGEMENT

SHIVAMOGGA – 577204

Information Science and Engineering

CERTIFICATE

This is to certify that the internship work carried out by **Megharaj P (4PM22IS029)** is a bonafide work of the student of the Bachelor of Engineering in Information Science and Engineering program of the institute, affiliated to Visvesvaraya Technological University, Belagavi.

The internship report entitled “**FoodShare: A Food Redistribution Web Application**” has been successfully completed over a duration of **Fifteen (15) Weeks** under the guidance of **Mr. Vinay SK, Assistant Professor, Department of Information Science and Engineering, PESITM, Shivamogga**. The report is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Information Science and Engineering.

It is further certified that all the corrections and suggestions indicated prior to the assessment and evaluation have been duly incorporated. The report has been approved as it satisfies the academic requirements prescribed by VTU, Belagavi and the institute for the award of the Bachelor of Engineering degree.

Internal Guide

Head of Department

Internship Completion Certificate

Scanned copy of the internship completion certificate can be placed here.

This placeholder is retained because the certificate image was not provided.

Acknowledgement

I express my sincere gratitude to everyone who supported me throughout the successful completion of my internship at Vstand4U Technologies Pvt. Ltd..

I am thankful to the management of PES Institute of Technology and Management, Shivamogga, for giving me this opportunity to undertake the internship as part of the academic curriculum.

I express my heartfelt gratitude to the Head of the Department of Information Science and Engineering for the encouragement and academic support extended during the internship period.

I am especially grateful to my internal guide, **Mr. Vinay SK, Assistant Professor, Department of Information Science and Engineering**, for his valuable guidance, motivation, and constructive suggestions during the internship and report preparation.

I also thank the mentors and technical team at Vstand4U Solutions for providing real-world exposure in full-stack development, system design, and deployment-oriented thinking through the FoodShare project.

Finally, I extend my thanks to my family and friends for their constant encouragement and moral support.

Megharaj P
4PM22IS029

Declaration

I hereby declare that the internship report entitled “**FoodShare: A Food Redistribution Web Application**” is a bonafide record of the work carried out by me during the academic year **2025–2026** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Information Science and Engineering**, under **Visvesvaraya Technological University, Belagavi**.

I further declare that this work has not been submitted previously, either in part or in full, to any other university or institution for the award of any degree or diploma.

Megharaj P

USN: 4PM22IS029

Place: Shivamogga

Date:

Vision and Mission of the Institution

Vision:

To be the most preferred institution for engineering and management education, research and entrepreneurship by creating professionally superior and ethically strong global manpower.

Mission:

- Provide high-quality engineering and management education with technical excellence for global requirements.
- Promote research and entrepreneurship with collaboration, modern facilities, and strong mentorship.
- Build ethically grounded graduates who succeed globally, lead with integrity and commit to lifelong learning and social responsibility.

Contents

Institute Certificate	i
Internship Completion Certificate	ii
Acknowledgement	iii
Declaration	iv
Vision and Mission of the Institution	v
1 Introduction	1
1.1 Overview of the Internship	1
1.2 Objectives of the Internship	1
1.3 Relevance to Sustainable Engineering	2
2 Sustainable Development Goals (SDGs)	3
2.1 Introduction to SDGs	3
2.2 SDGs Addressed During the Internship	3
2.3 Role of Computer Science in Achieving SDGs	4
3 Organization Profile	5
3.1 About Vstand4U Solutions	5
3.2 Vision and Mission	5
3.2.1 Vision	5
3.2.2 Mission	5
3.3 Training Environment	6

4	Problem Statement and Requirement Analysis	7
4.1	Problem Statement	7
4.2	Project Goal	7
4.3	Functional Requirements	7
4.4	Technology Stack	8
5	System Design and Architecture	9
5.1	Overall Architecture	9
5.2	System Modules	9
5.3	Database Design	10
5.4	Entity Relationship Flow	10
5.5	API Design	11
5.6	Application Workflow	12
5.7	Frontend Design Considerations	12
5.8	Backend Design Considerations	13
6	Implementation Details	14
6.1	Frontend Implementation	14
6.2	Backend Implementation	14
6.3	Database Initialization	15
6.4	Notification Handling	15
6.5	Deployment Readiness and Documentation	15
7	Internship Diary	16
8	Technical Work	18
8.1	Project Title: FoodShare	18
8.2	Core Features	18
8.3	Importance of the Chosen Architecture	18
8.4	Expected Project Directory Structure	19
8.5	Advantages of the Project	19
8.6	Challenges and Considerations	19

8.7 Conclusion of Technical Work	20
9 Learning Outcomes	21
10 Program Outcomes (POs) Addressed	22
11 Student Reflection	23
12 Conclusion	24
13 References	25
14 Annexure	26
Attendance and Topic Summary	26
Recommended Deliverables for the Actual Project Repository	26

List of Figures

5.1	High-Level Architecture of the FoodShare System	9
5.2	Logical Entity Relationship Flow in FoodShare	10
5.3	Functional Workflow of the FoodShare Application	12
8.1	Expected Project Directory Structure for FoodShare	19

List of Tables

4.1	Technology Stack Used for FoodShare	8
5.1	Database Schema Overview	10
5.2	Core API Endpoints	11
10.1	Program Outcomes Mapping	22
14.1	15-Week Attendance Report	26

Chapter 1

Introduction

1.1 Overview of the Internship

The internship at Vstand4U Solutions provided practical exposure to industry-oriented software development through the design of a socially relevant full-stack web application called FoodShare. The project focused on food redistribution, where surplus food from hotels, hostels, and related donors can be redirected to NGOs and coordinated with volunteers for pickup and delivery.

The internship offered an opportunity to understand the full software development lifecycle, including problem identification, requirement analysis, database design, API planning, user interface creation, testing, deployment readiness, and technical documentation. It helped bridge the gap between classroom knowledge and real-world development practices by combining frontend, backend, database, maps, and notification modules into one integrated system.

The project was intentionally selected for social impact. Food wastage is a major concern in urban and semi-urban areas, while many communities still face food insecurity. FoodShare addresses this gap by creating a digital bridge between food donors, volunteers, and NGOs. This made the internship technically rich as well as socially meaningful.

1.2 Objectives of the Internship

The major objectives of the internship were:

- To understand the planning and development of a complete full-stack web application.
- To gain practical exposure to React.js using functional components and hooks.

- To implement backend logic using Java, Servlets/controllers, and JDBC-based DAO patterns.
- To design and normalize a MySQL schema for application data storage.
- To integrate location-based features using Leaflet.js and OpenStreetMap.
- To understand CORS handling between a React frontend running on port 3000 and a Java backend running on port 8080.
- To explore notifications using NodeMailer or JavaMail API for pickup and donation alerts.
- To improve technical writing, system design, testing, and documentation skills.

1.3 Relevance to Sustainable Engineering

This internship strongly aligns with sustainable engineering principles because it focuses on efficient resource utilization, reduction of food waste, better coordination of logistics, and creation of a reusable digital platform for social benefit. Through FoodShare, technology is used not merely for automation, but for meaningful redistribution of excess resources in a responsible and scalable way.

Chapter 2

Sustainable Development Goals (SDGs)

2.1 Introduction to SDGs

The Sustainable Development Goals (SDGs) provide a global roadmap for solving key social, economic, and environmental challenges. Software systems can contribute significantly to these goals by improving coordination, accessibility, transparency, and decision-making. The FoodShare platform demonstrates how web technology can address a community-level problem in a practical way.

2.2 SDGs Addressed During the Internship

The internship and the FoodShare project align with the following SDGs:

▶ **SDG 2: Zero Hunger**

The core objective of FoodShare is to redistribute surplus food to those who need it, thereby directly contributing to hunger reduction.

▶ **SDG 3: Good Health and Well-Being**

The project includes food quantity, freshness, and expiry details so that safe and timely distribution decisions can be made.

▶ **SDG 11: Sustainable Cities and Communities**

By connecting donors, NGOs, and volunteers through a city-scale digital workflow, the system supports more organized and resilient community support networks.

▶ **SDG 12: Responsible Consumption and Production**

The application reduces avoidable food wastage by redirecting usable surplus food before expiry.

▷ **SDG 17: Partnerships for the Goals**

The project depends on collaboration among donors, volunteers, and NGOs, reflecting partnership-driven community impact.

2.3 Role of Computer Science in Achieving SDGs

Computer Science enables impactful SDG-oriented solutions through:

- development of user-friendly web interfaces,
- efficient backend workflows for data validation and storage,
- location-aware services for faster pickups,
- automated notifications for coordination, and
- reporting mechanisms that improve transparency and response time.

The internship demonstrated that well-designed software systems can create measurable value when they are aligned with a genuine social need.

Chapter 3

Organization Profile

3.1 About Vstand4U Solutions

Vstand4U Solutions is a technology-driven learning and development organization that emphasizes practical training, project-based learning, and career readiness. It provides students with hands-on exposure to current software technologies and industry-relevant workflows. The focus is not limited to syntax-level learning; instead, it encourages students to understand how systems are planned, developed, and documented in professional settings.

3.2 Vision and Mission

3.2.1 Vision

To create technically capable and professionally confident learners who can use modern technologies to solve real-world problems.

3.2.2 Mission

- To provide practical, industry-oriented technical training.
- To bridge academic concepts with real implementation.
- To encourage innovation, social awareness, and professional growth.
- To prepare students for software roles through projects and guided mentoring.

3.3 Training Environment

The internship environment encouraged structured learning, iterative improvement, and applied development. Regular technical guidance, design discussion, and implementation review helped shape the FoodShare project from concept to report-ready form. The organization also promoted documentation discipline, which is an important part of professional engineering practice.

Chapter 4

Problem Statement and Requirement Analysis

4.1 Problem Statement

Large quantities of usable food are often wasted by hotels, hostels, mess facilities, and event organizers, while many NGOs and communities continue to face shortages. Manual coordination for food redistribution is slow, unstructured, and unreliable. There is a need for a digital platform that can connect food donors with NGOs and volunteers in a timely and organized way.

4.2 Project Goal

The goal of FoodShare is to build a full-stack web application for food redistribution. The system should allow donors to publish available food, NGOs to view and request suitable donations, and volunteers to manage pickups and delivery logistics using location-based support.

4.3 Functional Requirements

- User registration and login with role selection: Donor, Volunteer, and NGO.
- Food donation form with quantity, expiry time, and status.
- NGO dashboard showing available food items.
- Pickup assignment and location tracking support.
- Map-based visualization using Leaflet.js and OpenStreetMap.

- Notification support for important status updates.
- REST endpoints for login, adding food, and fetching food records.

4.4 Technology Stack

Table 4.1: Technology Stack Used for FoodShare

Layer	Technology
Frontend	React.js using functional components and hooks
Backend	Java with JDBC, DAO pattern, Servlets or controllers
Database	MySQL
Maps	Leaflet.js with OpenStreetMap
Notifications	NodeMailer microservice or JavaMail API
Styling	Tailwind CSS or standard responsive CSS
Build / Config	Maven or Gradle, package.json, README, SQL initialization

Chapter 5

System Design and Architecture

5.1 Overall Architecture

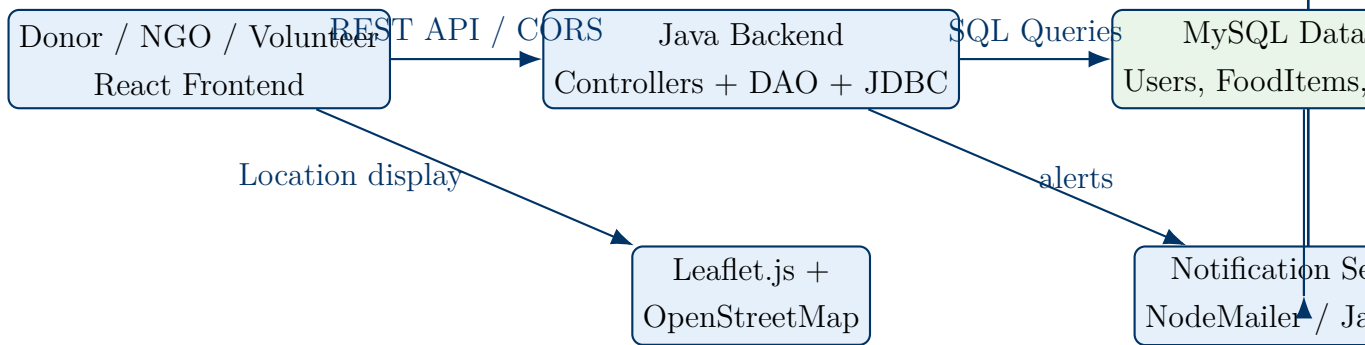


Figure 5.1: High-Level Architecture of the FoodShare System

5.2 System Modules

- **Authentication Module:** Handles login and signup with role selection.
- **Donation Module:** Allows donors to add food quantity, expiry time, and item status.
- **NGO Dashboard Module:** Displays available food records for verification and claiming.
- **Pickup Module:** Stores pickup assignments, latitude and longitude coordinates, and volunteer references.
- **Map Module:** Displays donation and pickup locations through markers.
- **Notification Module:** Sends alerts when food is added, claimed, or assigned.

5.3 Database Design

Table 5.1: Database Schema Overview

Table	Important Attributes
Users	user_id, name, email, password, role (Donor / Volunteer / NGO), phone
FoodItems	food_id, food_name, quantity, expiry_time, status, donor_id, created_at
Pickups	pickup_id, food_id, volunteer_id, ngo_id, location_lat, location_long, pickup_status

5.4 Entity Relationship Flow

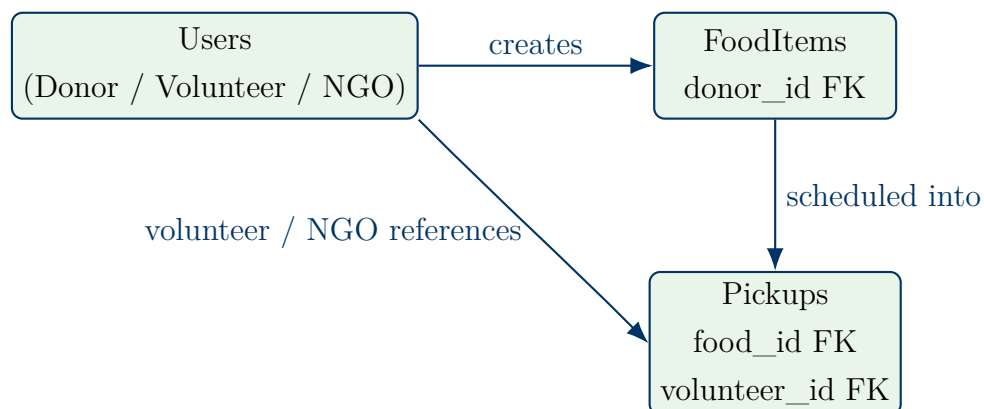


Figure 5.2: Logical Entity Relationship Flow in FoodShare

5.5 API Design

Table 5.2: Core API Endpoints

Endpoint	Method	Purpose
/api/login	POST	Validates credentials and role-based access
/api/signup	POST	Registers a new donor, NGO, or volunteer
/api/add-food	POST	Stores food donation details in the database
/api/get-food	GET	Returns available food records for dashboard display
/api/pickups	POST / GET	Creates or retrieves pickup assignments

5.6 Application Workflow

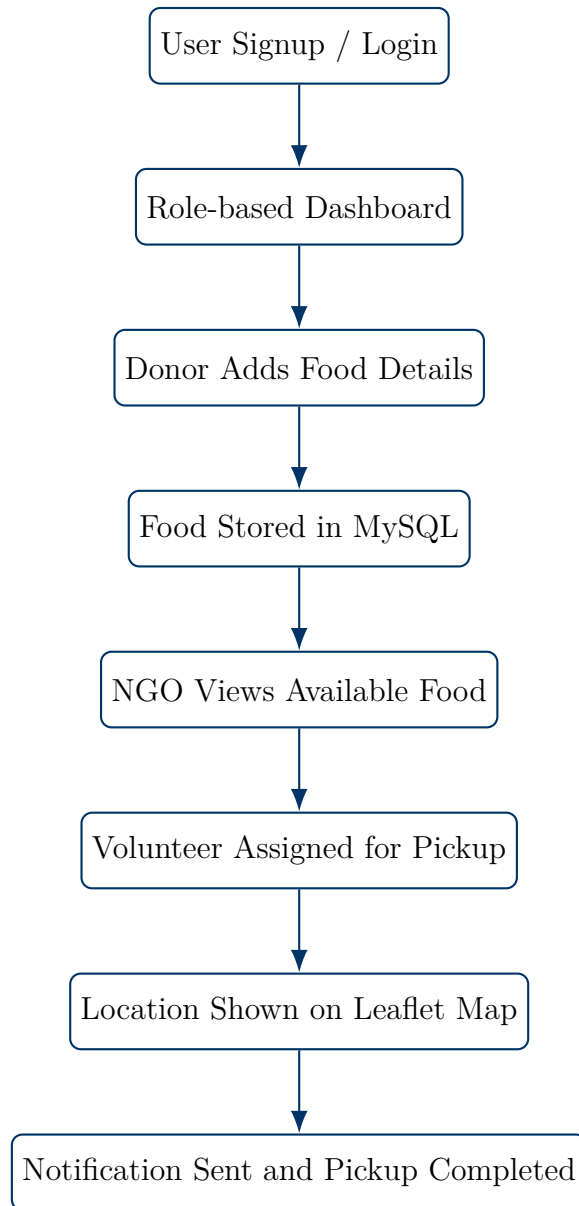


Figure 5.3: Functional Workflow of the FoodShare Application

5.7 Frontend Design Considerations

The frontend is designed using React.js with functional components and hooks. Key screens include login, signup, donor form, NGO dashboard, and map-based pickup view. The interface is intended to be clean, responsive, and impact-driven so that different user roles can use the application without confusion.

5.8 Backend Design Considerations

The backend follows a structured Java architecture with JDBC-based DAO classes. DAO separation improves maintainability and keeps database access logic independent from request handling. REST-style controllers or servlets process requests and return data to the frontend in a structured manner.

Chapter 6

Implementation Details

6.1 Frontend Implementation

The React frontend includes:

- Login and signup pages with role selection.
- Donate Food form for donors such as hotels and hostels.
- NGO dashboard displaying food item records in list form.
- React-Leaflet integration for pickup location markers.
- Responsive styling with a simple and impact-oriented visual hierarchy.

6.2 Backend Implementation

The Java backend includes:

- Maven or Gradle project structure.
- Database connection utility classes.
- DAO classes for `Users`, `FoodItems`, and `Pickups`.
- Controller or Servlet handling for `/api/login`, `/api/add-food`, and `/api/get-food`.
- CORS support so the React application on port 3000 can communicate with the backend on port 8080.

6.3 Database Initialization

The database schema is designed for role-based access and food tracking. Foreign key relationships ensure integrity between donors, food items, and pickup assignments. SQL initialization scripts are part of the expected project deliverables for clean setup.

6.4 Notification Handling

To support pickup coordination, notifications can be implemented using either:

- a lightweight Node.js microservice using NodeMailer, or
- JavaMail API directly from the backend.

This flexibility allows the system to adapt based on hosting convenience and deployment strategy.

6.5 Deployment Readiness and Documentation

The project also requires supporting files such as `pom.xml`, `package.json`, and `README.md`. These files are important because they make the project reproducible, portable, and easier for evaluators or collaborators to set up.

Chapter 7

Internship Diary

Week	Activities and Learning
Week 1	Understood the internship workflow, selected the social problem domain, and discussed the idea of a food redistribution platform. Learned how software requirements are derived from real-world community issues.
Week 2	Studied the functional requirements of FoodShare and identified the roles of donor, volunteer, and NGO. Prepared the initial feature list and workflow outline.
Week 3	Designed the basic MySQL schema for users, food items, and pickups. Learned the importance of foreign keys, role-based data modeling, and normalized structure.
Week 4	Planned the Java backend architecture with JDBC and DAO patterns. Understood separation between controllers, services, and data access logic.
Week 5	Worked on authentication flow and role-based login logic. Improved understanding of request handling and response design in web applications.
Week 6	Designed the Donate Food form and donor-side workflow. Learned how to capture quantity, expiry time, and food status meaningfully for real-world use.
Week 7	Implemented NGO dashboard planning and food-list retrieval flow. Focused on clear data presentation and role-based dashboard behavior.
Week 8	Explored React functional components and hooks for managing application state. Strengthened frontend structuring skills for reusable components.
Week 9	Integrated map planning using Leaflet.js and OpenStreetMap. Learned how free mapping solutions can support location-aware applications effectively.
Week 10	Designed the pickup assignment flow and volunteer coordination logic. Improved understanding of operational workflows within a full-stack system.

Week	Activities and Learning
Week 11	Studied notification integration using NodeMailer or JavaMail API. Learned how alerts can improve responsiveness in service-oriented applications.
Week 12	Focused on CORS handling between React on port 3000 and Java backend on port 8080. Understood frontend-backend communication constraints.
Week 13	Prepared system diagrams, API mapping, and module descriptions. Improved technical documentation and architectural presentation skills.
Week 14	Reviewed implementation strategy, database setup instructions, and supporting configuration files such as README, package.json, and pom.xml.
Week 15	Consolidated the complete report, refined technical explanations, and connected the project to sustainability and social impact outcomes.

Chapter 8

Technical Work

8.1 Project Title: FoodShare

FoodShare is a full-stack food redistribution web application built to connect donors, NGOs, and volunteers. It is designed to reduce food waste and improve the speed and transparency of food sharing workflows.

8.2 Core Features

- Role-based login and signup for Donor, Volunteer, and NGO.
- Donation form with quantity, expiry time, and food status.
- NGO dashboard for viewing available food items.
- Pickup tracking with latitude and longitude coordinates.
- Interactive map using React-Leaflet.
- Notification support for food addition and pickup events.
- REST communication between frontend and backend.

8.3 Importance of the Chosen Architecture

The architecture is suitable for learning and demonstration because each technology serves a clear purpose:

- React.js provides modular, responsive frontend interaction.
- Java and JDBC offer explicit backend logic and database connectivity.

- MySQL ensures structured persistence of user and food data.
- Leaflet.js enables free and practical map integration without paid APIs.
- NodeMailer or JavaMail supports event-driven coordination.

8.4 Expected Project Directory Structure

```
FoodShare/  
  frontend/  
    src/  
      components/  
      pages/  
      services/  
    package.json  
  backend/  
    src/main/java/...  
    src/main/webapp/  
    pom.xml  
  database/  
    foodshare.sql  
  README.md
```

Figure 8.1: Expected Project Directory Structure for FoodShare

8.5 Advantages of the Project

- Reduces wastage of usable food.
- Creates a structured communication channel among stakeholders.
- Encourages community participation and volunteer engagement.
- Uses accessible and mostly free technologies.
- Can be extended into a real civic-tech or NGO-support application.

8.6 Challenges and Considerations

- Accurate validation of food freshness and expiry data.
- Handling role-based access cleanly across modules.

- Reliable volunteer coordination and status updates.
- Maintaining secure authentication and input validation.
- Ensuring smooth API communication and deployment compatibility.

8.7 Conclusion of Technical Work

The FoodShare project demonstrates how full-stack development can be applied to a meaningful social problem. It integrates frontend development, backend architecture, database design, location services, and notification workflows into one cohesive application model.

Chapter 9

Learning Outcomes

- Understood how to convert a social problem statement into a software solution.
- Gained clarity in full-stack architecture involving React, Java, JDBC, and MySQL.
- Learned the purpose and use of DAO patterns in backend application design.
- Improved database design skills through role-based schema planning.
- Learned how interactive maps can be used in logistics and service applications.
- Strengthened documentation, workflow design, and requirement analysis skills.
- Improved technical confidence in integrating multiple technologies into one project.
- Understood how software projects can contribute to sustainability and social impact.

Chapter 10

Program Outcomes (POs) Addressed

Table 10.1: Program Outcomes Mapping

PO Code	Contribution
PO 1	Applied engineering knowledge in web application planning, database design, backend logic, and role-based workflows.
PO 2	Identified the real-world problem of food wastage and analyzed how software can enable efficient redistribution.
PO 3	Designed a complete solution architecture for FoodShare including frontend, backend, database, map, and notification modules.
PO 5	Used modern tools and technologies such as React.js, Java, JDBC, MySQL, Leaflet.js, and notification services.
PO 9	Improved communication and coordination skills by structuring the application around multi-stakeholder workflows.
PO 11	Developed independent learning and adaptability while exploring multiple technologies across the full-stack pipeline.

Chapter 11

Student Reflection

- **Challenges:** One major challenge was structuring a complete full-stack system around a real-world problem while keeping the design practical and scalable. Understanding how frontend interaction, backend APIs, database integrity, maps, and notifications fit together required careful planning.
- **Skills Gained:** The internship improved my confidence in system architecture, database design, role-based application workflows, React frontend planning, and Java backend thinking. It also strengthened my ability to prepare a technically organized report.
- **Improvements Needed:** I would like to deepen my practical implementation skills in backend security, deployment, real email notification setup, and advanced state handling in frontend applications.

Chapter 12

Conclusion

The internship at Vstand4U Technologies Pvt. Ltd. was a valuable academic and technical experience. Through the FoodShare project, I was able to understand how software engineering concepts can be combined to solve a meaningful societal issue. The project brought together full-stack design, structured data handling, API planning, mapping, and workflow coordination in one coherent application idea.

More importantly, the internship helped me realize that software projects can create tangible community impact when designed with purpose. FoodShare is a strong example of how technology can reduce waste, improve coordination, and support social good.

Chapter 13

References

- Vstand4U Solutions official website: <https://vstand4usolutions.com/>
- React.js documentation: <https://react.dev/>
- MySQL documentation: <https://dev.mysql.com/doc/>
- Leaflet.js documentation: <https://leafletjs.com/>
- OpenStreetMap project: <https://www.openstreetmap.org/>
- Java JDBC documentation: <https://docs.oracle.com/javase/tutorial/jdbc/>
- NodeMailer documentation: <https://nodemailer.com/>

Chapter 14

Annexure

Attendance and Topic Summary

Table 14.1: 15-Week Attendance Report

Timeline	Topic	Remark
Week 1	Internship orientation and problem selection	Present
Weeks 2–3	Requirement analysis and database schema planning	Present
Weeks 4–5	Java backend architecture and authentication flow	Present
Weeks 6–8	React frontend planning and dashboard design	Present
Weeks 9–10	Leaflet map integration and pickup workflow	Present
Weeks 11–12	Notification handling and CORS coordination	Present
Weeks 13–15	Documentation, diagrams, and final report preparation	Present

Recommended Deliverables for the Actual Project Repository

- `README.md` with setup instructions.
- SQL file for database initialization.
- `pom.xml` for Java backend configuration.
- `package.json` for React frontend dependencies.

- Source directories for frontend, backend, and database scripts.
- CORS configuration and environment notes for local execution.